

Value-Based Digital Health Innovation Canvas (VDHIC)

Instruction Guide

From Problem to Impact: Navigating Indonesia's Digital Health Ecosystem

Featuring the Quintuple Aim Framework

Population Health • Patient Experience • Provider Satisfaction • Cost Reduction • Health Equity

With Comprehensive Governance Framework

Regulatory • Clinical • Data • Technology

Australia Awards Fellowship 2025
Monash University • Digital Health Governance & Innovation

Version 1.2

1. Introduction

The Value-Based Digital Health Innovation Canvas (VDHIC) transforms the Indonesia Digital Health Transformation Toolkit conceptual framework into a practical, actionable tool. Inspired by the Business Model Canvas, VDHIC guides innovators through a structured journey from identifying health problems to delivering measurable value-based outcomes.

This canvas is designed specifically for the Indonesian digital health ecosystem, incorporating requirements for SATUSEHAT integration, regulatory sandbox participation, and alignment with national health priorities. The canvas anchors all innovation efforts to the **Quintuple Aim** framework, ensuring comprehensive value creation across all healthcare stakeholders.

Purpose

- Provide a structured framework for digital health innovation in Indonesia
- Ensure governance considerations (regulatory, clinical, data, and technology) are embedded from the start
- Facilitate multi-stakeholder dialogue with shared vocabulary
- Bridge the gap between conceptual design and practical implementation
- Support regulatory sandbox applications and investor pitches
- Align innovation with Quintuple Aim outcomes for comprehensive value creation

Target Users

- Digital health startups and entrepreneurs
- Healthcare organizations exploring digital transformation
- Accelerator and incubator programs
- Regulatory sandbox applicants
- Investors evaluating digital health opportunities
- Policy makers and health system planners

2. Canvas Structure

The VDHIC consists of four integrated layers that correspond to the original toolkit framework:

Governance Spine (Safety): Four governance blocks forming the comprehensive compliance framework

Main Canvas Grid: Nine blocks organized in three phases (Discover → Design → Deliver)

Value Anchor (Impact): Five Quintuple Aim outcome indicators

Governance Spine Components

Block	Focus	Key Elements
Regulatory Pathway	Compliance with laws and standards	BPOM, Sandbox, TKDN, ISO 13485
Clinical Governance	Patient safety and clinical quality	Protocols, PNPK, adverse events
Data Governance	Privacy and data protection	UU PDP, PP 71/2019, consent
Technology Governance	System reliability and security	Architecture, cybersecurity, DR/BC

Innovation Phases

Phase	Blocks	Focus
DISCOVER	1, 2, 3	Problem identification, user research, stakeholder mapping
DESIGN	4, 5, 6	Solution architecture, UX design, system integration
DELIVER	7, 8, 9	Evidence generation, scaling, sustainability

3. Block-by-Block Guide

Governance Spine

The governance spine forms the compliance umbrella over all innovation activities. These four blocks ensure safety, compliance, and technical reliability are considered from the outset, not as an afterthought.

Block 10: Regulatory Pathway

Key Questions:

- What regulatory classification applies (BPOM medical device, wellness app)?
- Is Regulatory Sandbox participation appropriate for your innovation?
- What TKDN (local content) requirements apply?
- What certifications are required (ISO 13485, IEC 62304)?

Block 11: Clinical Governance

Key Questions:

- What clinical protocols govern the intervention?
- How will clinical oversight be maintained?
- What adverse event reporting mechanisms exist?
- How does this align with PNPK (clinical guidelines)?

Block 12: Data Governance

Key Questions:

- How will you ensure UU PDP and PP 71/2019 compliance?
- What data minimization principles apply?
- How will consent be obtained and managed?
- What data retention and deletion policies apply?

Block 13: Technology Governance (NEW)

Technology governance ensures that the digital health solution is built on a reliable, secure, and maintainable technical foundation. This block addresses the infrastructure and systems aspects that are critical for operational success.

Key Questions:

- What is the system architecture (cloud, on-premise, hybrid)?
- What cybersecurity measures and standards apply (ISO 27001, encryption)?
- What API standards are used for system integration?
- What is the disaster recovery and business continuity plan (DR/BC)?
- How will version control and change management be handled?
- What is the system uptime target and SLA?
- How will technical debt be managed over time?

Indonesia-Specific Considerations:

- Compliance with PP 71/2019 on Electronic System and Transaction Implementation
- Data localization requirements for health data
- Connectivity resilience for areas with limited internet infrastructure
- Integration requirements with national systems (SATUSEHAT, BPJS)

DISCOVER Phase

The Discover phase focuses on deeply understanding the problem space before jumping to solutions.

Block 1: Health Challenge

- What specific health problem are you solving?
- What is the current patient journey and pain points?
- How does this align with Renstra Kemenkes/SDGs?

Block 2: User Profiles

- Primary users: Patients, Providers, or Administrators?
- What are their behaviors, motivations, and constraints?
- Indonesia context: Urban/rural divide, BPJS segments?

Block 3: Stakeholder Map

- Government, Clinical, Payer, Industry, and Academic stakeholders?
- Who are the champions and potential blockers?

DESIGN Phase

The Design phase translates insights into concrete solutions.

Block 4: Solution Architecture

- Which pillar: Digital Medicine, Diagnostics, Therapeutics, or Wellness?
- What risk classification applies (Class I, II, III)?

Block 5: User Experience

- How will you ensure usability (Nielsen heuristics, SUS score target)?
- Accessibility: Bahasa Indonesia, offline mode?

Block 6: SATUSEHAT Integration

- Which FHIR resources will you implement?
- Technical readiness for national interoperability?

DELIVER Phase

The Deliver phase focuses on validation, scaling, and sustainability.

Block 7: Evidence Plan

- What is your evidence pathway: Pilot → RCT → Real-World Evidence?
- What clinical endpoints will demonstrate efficacy?

Block 8: Scale Strategy

- What change management approach for provider adoption?
- How will you scale from pilot to national deployment?

Block 9: Business Model

- What revenue model: B2B, B2C, or B2G?
- Potential BPJS reimbursement pathway (INACBGs, kapitasi)?

Value Anchor: Quintuple Aim

Every innovation must demonstrate clear alignment with the Quintuple Aim outcomes.

1. Population Health

What clinical outcomes will improve at the population level?

2. Patient Experience

How will patient satisfaction and engagement increase?

3. Provider Satisfaction

How will this improve clinician workflow and reduce burden?

4. Cost Reduction

What efficiency gains are expected?

5. Health Equity

How does this reduce healthcare disparities?

4. How to Use the Canvas

Workshop Format (2.5-3 hours)

1. **Introduction (15 min):** Overview of VDHIC framework and Quintuple Aim
2. **Discover Phase (45 min):** Work through Blocks 1-3
3. **Design Phase (45 min):** Complete Blocks 4-6
4. **Deliver Phase (30 min):** Outline Blocks 7-9
5. **Governance Review (25 min):** Complete Blocks 10-13 (including Technology Governance)
6. **Value Alignment (15 min):** Map outcomes to all five Quintuple Aim dimensions
7. **Wrap-up (10 min):** Identify gaps and next steps

Maturity Assessment

Use maturity indicators to track progress for each block:

- ○ **Concept:** Hypothesis only, no validation yet
- ● **Validated:** Evidence exists, assumptions tested
- ● **Implemented:** Deployed and operational
- ★ **Scaled:** National reach or significant market penetration

About

The Value-Based Digital Health Innovation Canvas (VDHIC) was developed as part of the Australia Awards Fellowship 2025 at Monash University. It transforms the Indonesia Digital Health Transformation Toolkit conceptual framework into a practical tool for innovators.

Governance Framework: The canvas incorporates four governance domains—Regulatory, Clinical, Data, and Technology—ensuring comprehensive oversight from the earliest stages of innovation design.

Quintuple Aim: The value anchor extends traditional outcome frameworks to include provider satisfaction and health equity as explicit goals.

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